INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP

DataCo Supply Chain Analytics

Consolidated Project Report — Milestones 1 to 4

Team Members

Vaishnavi

Shreya

Pujitha

Siva Reddy

Imam Khasim

"From Data Cleaning to Actionable Supply Chain Insights"

Executive Summary

This report consolidates the full DataCo Supply Chain Analytics project completed as part of the Infosys Springboard Virtual Internship. The project progressed through four milestones — data preprocessing, exploratory data analysis and visualization, business-insight generation, and interactive Power BI dashboard development — transforming a raw, 2,599-row operational dataset into a decision-support analytics solution covering sales, delivery, inventory, and regional performance.

Across all milestones, the dataset consistently reflects the following headline metrics:

Total Sales	**₹6.13M**
Total Orders	**3,000 (3K)**
Average Delivery Days	**3.95 days**
On-Time Delivery Rate	**32.13%**
Total Inventory	**40,000 units (40K)**
Inventory Value	**₹11.65M**
Total Profit	**₹220.13K**

The single most significant finding across the project is that on-time delivery performance, at roughly 32%, is well below an acceptable threshold and represents the primary opportunity for operational improvement identified by the team.

Milestone 1: Data Preprocessing

Introduction and Why Preprocessing Matters

Supply chain management involves coordinating the flow of goods, information, and finances across suppliers, manufacturers, and customers to ensure timely and cost-effective delivery. Reliable supply chain decisions depend on clean, accurate, and consistent data — missing or inconsistent records can distort analysis and mislead conclusions. Raw operational data typically contains gaps, duplicates, and formatting issues, so preprocessing was carried out first to ensure the dataset was trustworthy before any deeper analysis began.

Objectives

1. Understand the dataset — explore structure, columns, and data types across all 24 fields.
2. Identify missing values — detect gaps across the dataset before they affect analysis.
3. Handle missing data — apply median/mode imputation for numeric and categorical fields.
4. Remove duplicate records — check and eliminate duplicate order entries.
5. Standardize the dataset — correct inconsistent category labels and column names.
6. Convert date columns — transform order and shipping dates into datetime format.
7. Prepare for analysis — deliver a clean dataset ready for further exploration.

Dataset Overview

Rows	**2,599**
Columns	**24**
Rows with Missing Values	**138**

The dataset is order-level operational data covering shipping performance, customer segments, product categories, sales, and delivery outcomes across five global markets: LATAM, USCA, Pacific Asia, Africa, and Europe. Major column groups include:

- Order ID, Type, Order Status — order identity and payment/status details
- Days for shipping (real / scheduled), Delivery Status, Late delivery risk — shipping performance
- Category Name, Product Name, Department Name — product classification
- Customer Segment, Customer City, Market, Order Region/Country — customer and geography
- Sales, Unit Price, Order Item Total, Order Profit Per Order — financial metrics
- Order date, shipping date, Shipping Mode, Inventory Stock (Units) — logistics detail

Preprocessing Steps (Python)

1. Loading & Inspection

Imported Pandas and NumPy, loaded the Excel dataset, and reviewed initial records, dimensions (2,599 × 24), and data types for every column.

2. Handling Missing Values

Filled numerical gaps (e.g., Sales, Order Item Total, Inventory Stock) using the column median, and categorical gaps (Category Name, Market, Shipping Mode) using the column mode.

3. Duplicates & Standardization

Checked for duplicate order records, converted order date and shipping date to datetime format, and standardized inconsistent category labels and column names.

Key Code Snippets

Step	Code
Loading & Inspection	import pandas as pd; import numpy as np df = pd.read_excel('dataco.xlsx') print(df.shape); print(df.dtypes)
Handling Missing Values	df['Sales'].fillna(df['Sales'].median(), inplace=True) df['Category Name'].fillna(df['Category Name'].mode()[0])
Duplicates & Standardization	df.drop_duplicates(inplace=True) df['order date'] = pd.to_datetime(df['order date'])

Preprocessing Outcome

- Missing values across numerical and categorical fields handled successfully.
- Duplicate records checked and confirmed absent.
- Data types corrected for accurate computation.
- Order and shipping dates converted to proper datetime format.
- Column names and category labels standardized.
- Dataset is now clean, consistent, and ready for further analysis.

Conclusion

Preprocessing transformed the raw DataCo supply chain data into a reliable, well-structured dataset. By resolving missing values, verifying duplicates, and correcting formats, the data became consistent and analysis-ready, laying a strong foundation for future exploratory and visual insights.

Milestone 2: Data Analysis & Visualization

With preprocessing complete, the cleaned DataCo dataset was used to build an interactive dashboard and perform visual analysis across the core areas of the supply chain: Clean Dataset → KPI Analysis → Sales Analysis → Delivery Analysis → Inventory Analysis → Regional Analysis → Business Insights.

Main areas analyzed: Sales, Orders, Delivery, Inventory, Products, Categories, Regions, and Cities.

Supply Chain Vision: End-to-End Dashboard

A centralized dashboard provided a single view of sales, orders, delivery performance, inventory, products, categories, regions, and cities.

Total Sales	**6.13M**
Total Orders	**3K**
Avg Delivery Days	**3.95**
On-Time %	**32.13%**
Total Inventory	**40K**
Inventory Value	**11.65M**

Sales Trend Analysis

The sales trend chart (Jan 2023 – Jan 2025) shows sales performance changing over time with periodic increases and decreases, giving an overview of overall sales movement. Business relevance: sales trends help identify periods of stronger or weaker business performance.

Delivery Status Analysis

Delivery status was tracked across three categories: Late delivery, Shipping on time, and Advance shipping. The distribution provides a quick view of how orders are progressing through the delivery process.

Key Takeaway

Monitoring delivery status helps identify areas where delivery performance requires attention.

Regional Sales Performance

Sales were compared across five regions — South, East, West, North, and Central.

Key Insight

Regional comparison makes it easier to identify stronger and weaker sales-performing regions. South emerged as the strongest displayed region by sales.

Regional Delivery Performance

Delivery-status categories (Advance shipping, Late delivery, Shipping on time) were compared across South, East, North, Central, and West regions.

Business Relevance

The visualization shows how delivery-status patterns differ across regions. Regional delivery analysis can help management identify locations where delivery performance needs closer monitoring.

Product, Category & City Insights

Additional views broke down performance further:

- Inventory by Product — top items include Monitor, Desk, Headphones, and Backpack.
- Sales by Category — Furniture, Home, and Electronics lead the category mix.
- Sales by City — geographic distribution of sales mapped by customer city.

Product-level inventory, category-level sales, and city-level distribution provide more detailed views of supply-chain performance.

Milestone 2 Conclusion

Conclusion

The project transformed the cleaned DataCo supply-chain dataset into a visual analytics solution that provides an end-to-end view of sales, orders, delivery, inventory, products, categories, regions, and cities.

Analytical Value

- Sales trends provide visibility into performance over time.
- Regional sales analysis highlights geographical differences.
- Delivery-status analysis supports monitoring of shipping performance.
- Product inventory analysis provides visibility into stock levels.
- Category and city analysis provide deeper business insights.

Milestone 3: Business Insights & Advanced Analysis

This milestone extended the analysis into supplier and regional scoreboards, shipping efficiency, route/carrier drill-downs, and advanced KPI heat maps, deepening the decision-support value of the dataset.

Supplier & Regional Performance Scoreboard

A combined view of sales, orders, inventory, and delivery performance across regions, with interactive filters for Order Country and Order Status.

Total Sales	**6.13M**
Total Orders	**3K**
Total Inventory	**11M**
On Time %	**32.13%**
Avg Delivery Days	**3.95**

Key Observations

- South is the strongest displayed region by sales.
- Performance varies across South, East, West, North, and Central regions.
- Overall on-time delivery is approximately 32.13%, an important improvement area.
- Shipping-mode performance can be compared to spot operational issues.
- Country and order-status filters allow deeper investigation.

Business Value

Helps management compare regional performance, monitor delivery reliability, and identify areas needing operational attention.

Transportation Cost & Shipping Efficiency Analysis

No direct transportation-cost field exists in the dataset, so efficiency was analyzed via shipping and delivery metrics instead: Shipping Mode, Actual vs. Scheduled Shipping Days, Delivery Status, Late Delivery Risk, and Order Region/Country.

The dataset contains 2,599 orders totaling approximately $6.13 million in sales and $220.13 thousand in total profit. Approximately 41.36% of orders carry a late delivery risk, indicating that delivery performance is an important area to monitor. Average shipping time is 3.95 days, and average inventory stock is approximately 4,061 units. Among delivery statuses, Late Delivery has the highest order count at 1,083 orders, and the Corporate customer segment has the highest order count at 925 orders.

Key Questions Explored

- Which shipping modes are more efficient?
- Which regions experience greater delivery risk?
- How does shipping performance vary by delivery status?
- Which logistics areas require improvement?

Business Insight

Shipping efficiency influences delivery reliability and operational performance. Comparing modes, regions, and outcomes reveals opportunities for logistics optimization.

Route & Carrier Performance Analysis

A drill-down view of routes, departments, shipping modes, and delivery outcomes across markets: Order Country → Department → Shipping Mode → Delivery Status.

Total Sales	**6.13M**
Inventory Value	**11M**
Late Delivery Risk	**1.08K**

Key Insights

- Route Visibility — country, department, shipping mode, and delivery combinations can be investigated.
- Delivery Risk — regional late-delivery risk can be identified and monitored.
- Shipping Efficiency — average shipping days can be compared across delivery statuses.
- Market Coverage — order distribution across markets provides geographic context.
- Operational Action — high-risk regions and route combinations can be prioritized.

Note: carrier-level performance is not claimed as best or worst unless actual carrier data is present in the source.

Advanced KPI Visualization

A heat map compared delivery performance across region, customer segment, and shipping mode (Order Region × Customer Segment × Shipping Mode), helping identify combinations linked to higher delivery time or risk.

KPI Insight Cards

- Delivery Speed — average delivery time is approximately 3.95 days.
- On-Time Performance — approximately 32.13% of orders are on time.
- Delivery Risk — late-delivery risk should be monitored by region and mode.
- Segment Performance — customer segments show different delivery patterns.

Sales & Order Analysis

Total Sales	**6.13M**
Total Orders	**3K**
Total Inventory	**40K**
Customer Segments	**Consumer | Corporate | Home Office**

Key Insights

- Sales Trend — performance varies across months and quarters.
- Regional Strength — South is the strongest displayed region by sales.
- Category Mix — Furniture and Home are among the leading displayed categories.
- Payment Types — sales are distributed across Debit, Payment, Transfer, and Cash.
- Customer Segment — filters allow comparison of Consumer, Corporate, and Home Office performance.

Business Value

Sales and order analysis provides visibility into revenue trends, regional performance, category contribution, and customer-segment behavior, supporting data-driven business decisions.

Inventory & Delivery Performance Analysis

This view combined inventory position with delivery and shipping performance across regions and shipping modes.

Total Inventory	**40K**
Inventory Value	**11.65M**
Avg Delivery Days	**3.95**
On-Time %	**32.13%**
Late Delivery Risk	**1.08K**
Total Orders	**3K**

Inventory Insights

- Inventory spans multiple product categories (~40K units).
- Inventory value is approximately 11.65M.
- Inventory value varies by shipping mode and customer city.

Delivery Insights

- Average delivery time is approximately 3.95 days.
- On-time delivery is approximately 32.13% — a key improvement area.
- Late-delivery risk varies across order regions.

Business Value

Combining inventory and delivery analysis helps management monitor stock position while identifying logistics areas that need attention.

Milestone 3 Key Insights & Conclusion

- Sales Performance — visibility into sales trends across time, regions, categories, and order types.
- Regional Performance — South is the strongest displayed sales region; regional comparisons highlight differences in business performance.
- Inventory Visibility — approximately 40K inventory units and 11.65M inventory value show the importance of effective monitoring.
- Delivery Performance — average delivery time is approximately 3.95 days and on-time delivery is approximately 32.13%, an important improvement area.
- Logistics & Risk — shipping-mode, delivery-status, and regional late-delivery-risk analysis adds visibility into logistics performance.
- Interactive Decision Support — Power BI dashboards allow users to explore sales, inventory, delivery, regions, categories, and other supply-chain dimensions interactively.

Conclusion

The project transformed the cleaned DataCo supply-chain dataset into an interactive visual analytics solution covering sales, orders, inventory, delivery performance, shipping modes, regions, categories, and operational risk. The dashboards provide a decision-support framework that helps identify performance gaps, monitor key supply-chain KPIs, and prioritize opportunities for operational improvement.

Milestone 4: Interactive Dashboard Development

The final milestone consolidated all prior analysis into four purpose-built, interactive Power BI dashboards, giving management an integrated view of supply-chain performance — from executive KPIs to operational and regional detail.

Solution Overview

Together, the four dashboards provide an integrated view of supply-chain performance — from management KPIs to operational and regional analysis.

#	Dashboard	Focus
1	Executive Overview	Overall supply-chain performance and management KPIs
2	Warehouse Efficiency	Orders, shipping efficiency, and delivery performance
3	Inventory & Product Performance	Inventory, products, sales, and profitability
4	Delivery & Regional Performance	Delivery reliability, markets, countries, regions, and shipping modes

Dashboard 1 — Executive Overview

Provides supply chain performance at a glance, filterable by order date, market, category, and order status.

Total Orders	**3K**
Total Sales	**6.13M**
Total Profit	**220.13K**
Avg Shipping Days	**3.95**

Business Value

Enables management to quickly monitor overall supply-chain performance and identify areas requiring attention.

Dashboard 2 — Warehouse Efficiency

Analyzes orders by month, average shipping days by month, delivery status distribution, late delivery risk by market, and average shipping days by shipping mode.

Total Orders	**3K**

Avg Shipping Days	3.95
Late Delivery Rate	42%

Business Value

Helps identify operational bottlenecks and opportunities to improve warehouse and shipping efficiency.

Dashboard 3 — Inventory & Product Performance

Covers sales by product category, average inventory by category, top 10 products by sales, top 10 products by profit, and total sales/profit by category.

Total Products	10
Average Inventory	4.06K
Total Sales	₹6.13M
Total Profit	₹220.13K
Total Quantity	14K

Business Value

Provides visibility into inventory position, product performance, category contribution, and profitability.

Dashboard 4 — Delivery & Regional Performance

Filterable by market, order region, order country, and shipping mode; analyzes late delivery rate by market, delivery performance by region, late delivery rate by country, actual vs. scheduled shipping days by market, and late delivery rate by shipping mode.

Total Orders	3K
Late Delivery Rate	41.7%
On-Time Delivery	32.1%
Avg Delivery Gap	0.45 days

Key Insight

On-time delivery at ~32% flags delivery reliability as a critical improvement area. Performance varies significantly across markets and regions.

From Data to Actionable Supply Chain Insights

The four interactive dashboards transform supply-chain data into a decision-support solution covering overall performance, warehouse efficiency, inventory and product performance, and delivery and regional performance.

Stage	Description
1. Data	Raw supply-chain records
2. Analysis	KPI extraction & modelling
3. Dashboards	4 interactive Power BI views
4. Insights	Actionable business decisions

Overall Project Conclusion

Across all four milestones, the DataCo Supply Chain Analytics project moved systematically from raw, inconsistent operational data to a polished, interactive analytics solution:

- Milestone 1 established a clean, trustworthy dataset through missing-value imputation, duplicate checks, date conversion, and standardization.
- Milestone 2 turned that clean dataset into exploratory visualizations covering sales trends, delivery status, and regional and product-level performance.
- Milestone 3 deepened the analysis with supplier/regional scoreboards, shipping-efficiency and route/carrier drill-downs, and advanced KPI heat maps.
- Milestone 4 consolidated everything into four interactive Power BI dashboards — Executive Overview, Warehouse Efficiency, Inventory & Product Performance, and Delivery & Regional Performance — giving stakeholders a decision-support framework across the full supply chain.

The consistent headline finding throughout the project is that on-time delivery performance (approximately 32%) is the most significant improvement opportunity, while the South region and the Furniture/Home categories consistently emerge as top performers by sales. Together, the milestones deliver on the project's guiding theme:

"From Data Cleaning to Actionable Supply Chain Insights"

Infosys Springboard Virtual Internship — Vaishnavi | Shreya | Pujitha | Siva Reddy | Imam Khasim